

Merged_Cleaned_wb

August 6, 2024

1 Independent Validation of a Merged and Cleaned Dataset

```
[ ]: # Standard package imports
```

```
# Data analysis
```

```
import pandas as pd
```

```
import numpy as np
```

```
# Plotting and Correlation Maths
```

```
import seaborn as sns
```

```
import scipy as sci
```

```
# Simple model development
```

```
import sklearn as sk
```

```
import os
```

```
[ ]: weather_df_allegheny = pd.read_csv('/Users/hunterberberich/Desktop/Data/  
↳Allegheny County.csv')
```

```
[ ]: print(weather_df_allegheny.info())
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 2678 entries, 0 to 2677
```

```
Data columns (total 33 columns):
```

#	Column	Non-Null Count	Dtype
0	name	2678 non-null	object
1	datetime	2678 non-null	object
2	tempmax	2678 non-null	float64
3	tempmin	2678 non-null	float64
4	temp	2678 non-null	float64
5	feelslikemax	2678 non-null	float64
6	feelslikemin	2678 non-null	float64
7	feelslike	2678 non-null	float64
8	dew	2678 non-null	float64
9	humidity	2678 non-null	float64
10	precip	2678 non-null	float64

```

11  precipprob      2678 non-null  int64
12  precipcover     2678 non-null  float64
13  preciptype      1648 non-null  object
14  snow            2678 non-null  float64
15  snowdepth       2678 non-null  float64
16  windgust        2678 non-null  float64
17  windspeed       2678 non-null  float64
18  winddir         2678 non-null  float64
19  sealevelpressure 2678 non-null  float64
20  cloudcover      2678 non-null  float64
21  visibility      2678 non-null  float64
22  solarradiation   2678 non-null  float64
23  solarenergy     2678 non-null  float64
24  uvindex         2678 non-null  int64
25  severerisk      843 non-null   float64
26  sunrise         2678 non-null  object
27  sunset          2678 non-null  object
28  moonphase       2678 non-null  float64
29  conditions      2678 non-null  object
30  description     2678 non-null  object
31  icon            2678 non-null  object
32  stations        2678 non-null  object

```

dtypes: float64(22), int64(2), object(9)

memory usage: 690.6+ KB

None

```
[ ]: print(weather_df_allegheny.describe())
```

	tempmax	tempmin	temp	feelslikemax	feelslikemin	\
count	2678.000000	2678.000000	2678.000000	2678.000000	2678.000000	
mean	61.264787	44.997125	52.911875	60.119828	41.265534	
std	18.413584	16.774329	17.246091	20.840241	20.475064	
min	8.200000	-5.100000	2.100000	-8.600000	-30.400000	
25%	46.500000	31.600000	38.900000	42.900000	24.600000	
50%	63.200000	45.400000	53.900000	63.200000	41.950000	
75%	77.600000	60.475000	68.700000	77.600000	60.475000	
max	93.700000	75.500000	83.300000	100.600000	75.500000	

	feelslike	dew	humidity	precip	precipprob	...	\
count	2678.000000	2678.000000	2678.000000	2678.000000	2678.000000	...	
mean	50.457506	41.792532	68.682450	0.122696	54.294249	...	
std	20.340899	17.727614	13.299889	0.273889	49.824556	...	
min	-16.500000	-10.800000	24.700000	0.000000	0.000000	...	
25%	34.100000	27.625000	59.700000	0.000000	0.000000	...	
50%	53.200000	42.300000	69.600000	0.003000	100.000000	...	
75%	68.700000	57.975000	78.300000	0.105750	100.000000	...	
max	87.200000	74.500000	99.200000	3.922000	100.000000	...	

	windspeed	winddir	sealevelpressure	cloudcover	visibility \
count	2678.000000	2678.000000	2678.000000	2678.000000	2678.000000
mean	12.731889	212.700560	1017.599552	54.781740	8.912211
std	4.562395	85.427759	6.650963	29.013607	1.339226
min	4.100000	0.000000	992.500000	0.000000	1.100000
25%	9.300000	162.325000	1013.400000	29.800000	8.500000
50%	12.200000	222.650000	1017.400000	55.250000	9.500000
75%	15.500000	276.275000	1021.800000	80.475000	9.900000
max	34.800000	359.900000	1042.000000	100.000000	9.900000

	solarradiation	solarenergy	uvindex	severerisk	moonphase
count	2678.000000	2678.000000	2678.000000	843.000000	2678.000000
mean	125.280358	10.809858	4.795370	14.827995	0.483342
std	79.305213	6.849332	2.471342	14.298733	0.288079
min	5.800000	0.500000	0.000000	10.000000	0.000000
25%	59.050000	5.100000	3.000000	10.000000	0.250000
50%	110.550000	9.500000	5.000000	10.000000	0.490000
75%	184.075000	15.900000	7.000000	10.000000	0.747500
max	345.400000	29.800000	10.000000	100.000000	0.980000

[8 rows x 24 columns]

```
[ ]: print(weather_df_allegHENY.isnull().sum())
```

name	0
datetime	0
tempmax	0
tempmin	0
temp	0
feelslikemax	0
feelslikemin	0
feelslike	0
dew	0
humidity	0
precip	0
precipprob	0
precipcover	0
preciptype	1030
snow	0
snowdepth	0
windgust	0
windspeed	0
winddir	0
sealevelpressure	0
cloudcover	0
visibility	0
solarradiation	0
solarenergy	0

```

uvindex          0
severerisk       1835
sunrise          0
sunset           0
moonphase        0
conditions        0
description       0
icon             0
stations         0
dtype: int64

```

```
[ ]: columns_to_remove = ['stations', 'sunrise', 'sunset', 'conditions',
    ↪ 'sealevelpressure', 'solarradiation', 'description', 'solarenergy', 'winddir',
    ↪ 'visibility', 'windspeed', 'windgust', 'precipprob']
```

```
[ ]: weather_df_allegheny2 = weather_df_allegheny.drop(columns=columns_to_remove)
```

```
[ ]: print(weather_df_allegheny2)
```

	name	datetime	tempmax	tempmin	temp	feelslikemax	\
0	Allegheny County	2017-01-01	43.9	31.4	37.1	40.1	
1	Allegheny County	2017-01-02	46.3	37.0	42.6	43.3	
2	Allegheny County	2017-01-03	48.6	42.9	45.1	46.3	
3	Allegheny County	2017-01-04	44.3	21.1	33.8	37.4	
4	Allegheny County	2017-01-05	22.6	19.0	20.6	20.8	
...	...	...	...	...	...	...	
2673	Allegheny County	2024-04-27	79.1	53.0	65.3	79.1	
2674	Allegheny County	2024-04-28	82.0	63.2	72.6	81.5	
2675	Allegheny County	2024-04-29	82.9	62.6	74.1	82.1	
2676	Allegheny County	2024-04-30	72.3	61.1	65.9	72.3	
2677	Allegheny County	2024-05-01	79.9	53.7	67.5	79.9	

	feelslikemin	feelslike	dew	humidity	precip	precipcover	preciptype	\
0	29.0	33.2	28.3	71.1	0.025	12.50	rain,snow	
1	33.2	38.4	38.9	87.1	0.095	29.17	rain	
2	37.7	41.6	42.6	91.0	0.793	70.83	rain	
3	8.0	23.6	23.4	67.4	0.052	29.17	rain,snow	
4	7.0	11.2	11.5	68.4	0.056	25.00	snow	
...	...	...	...	...	...	...	...	
2673	53.0	65.3	50.3	60.5	0.090	25.00	rain	
2674	63.2	72.6	55.3	55.6	0.000	0.00	rain	
2675	62.6	73.9	54.6	52.0	0.000	0.00	rain	
2676	61.1	65.9	56.1	71.0	0.014	4.17	rain	
2677	53.7	67.5	48.6	55.4	0.000	0.00	NaN	

	snow	snowdepth	cloudcover	uvindex	severerisk	moonphase	\
0	0.0	0.2	41.7	4	NaN	0.12	
1	0.0	0.0	87.9	1	NaN	0.16	

2	0.0	0.0	78.7	0	NaN	0.19
3	0.0	0.0	73.0	4	NaN	0.23
4	0.2	0.0	83.5	3	NaN	0.25
...	...	...	...	...	...	...
2673	0.0	0.0	65.9	7	10.0	0.63
2674	0.0	0.0	47.9	10	10.0	0.67
2675	0.0	0.0	27.2	9	10.0	0.70
2676	0.0	0.0	68.1	6	10.0	0.74
2677	0.0	0.0	16.2	8	10.0	0.75

	icon
0	rain
1	rain
2	rain
3	rain
4	snow
...	...
2673	rain
2674	partly-cloudy-day
2675	partly-cloudy-day
2676	rain
2677	clear-day

[2678 rows x 20 columns]

```
[ ]: weather_df_alleggheny2['preciptype'] = weather_df_alleggheny2['preciptype'].
      ↪fillna('none')
```

```
[ ]: weather_df_alleggheny2['severerisk'] = weather_df_alleggheny2['severerisk'].
      ↪fillna(0)
      print(weather_df_alleggheny2['severerisk'])
```

0	0.0
1	0.0
2	0.0
3	0.0
4	0.0

...	
2673	10.0
2674	10.0
2675	10.0
2676	10.0
2677	10.0

Name: severerisk, Length: 2678, dtype: float64

```
[ ]: weather_df_alleggheny2['icon'] = weather_df_alleggheny2['icon'].str.
      ↪replace('-day', '', regex=False)
```

```
unique_values = weather_df_allegheny2['icon'].unique().tolist()
print(unique_values)
```

```
['rain', 'snow', 'partly-cloudy', 'clear', 'cloudy']
```

```
[ ]: weather_df_allegheny2['datetime'] = pd.
      ↳to_datetime(weather_df_allegheny2['datetime'])
```

```
[ ]: print(weather_df_allegheny2.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2678 entries, 0 to 2677
Data columns (total 20 columns):
#   Column                Non-Null Count  Dtype
---  -
0   name                  2678 non-null  object
1   datetime              2678 non-null  datetime64[ns]
2   tempmax               2678 non-null  float64
3   tempmin               2678 non-null  float64
4   temp                  2678 non-null  float64
5   feelslikemax          2678 non-null  float64
6   feelslikemin          2678 non-null  float64
7   feelslike             2678 non-null  float64
8   dew                   2678 non-null  float64
9   humidity              2678 non-null  float64
10  precip                2678 non-null  float64
11  precipcover           2678 non-null  float64
12  preciptype            2678 non-null  object
13  snow                  2678 non-null  float64
14  snowdepth             2678 non-null  float64
15  cloudcover            2678 non-null  float64
16  uvindex               2678 non-null  int64
17  severerisk            2678 non-null  float64
18  moonphase             2678 non-null  float64
19  icon                  2678 non-null  object
dtypes: datetime64[ns](1), float64(15), int64(1), object(3)
memory usage: 418.6+ KB
None
```

```
[ ]: weather_df_allegheny2
```

```
[ ]:
      name      datetime  tempmax  tempmin  temp  feelslikemax  \
0  Allegheny County  2017-01-01    43.9     31.4   37.1         40.1
1  Allegheny County  2017-01-02    46.3     37.0   42.6         43.3
2  Allegheny County  2017-01-03    48.6     42.9   45.1         46.3
3  Allegheny County  2017-01-04    44.3     21.1   33.8         37.4
4  Allegheny County  2017-01-05    22.6     19.0   20.6         20.8
```

...	...	...	...	...	...	...	...	...
2673	Allegheny County	2024-04-27	79.1	53.0	65.3	79.1		
2674	Allegheny County	2024-04-28	82.0	63.2	72.6	81.5		
2675	Allegheny County	2024-04-29	82.9	62.6	74.1	82.1		
2676	Allegheny County	2024-04-30	72.3	61.1	65.9	72.3		
2677	Allegheny County	2024-05-01	79.9	53.7	67.5	79.9		

	feelslikemin	feelslike	dew	humidity	precip	precipcover	preciptype	\
0	29.0	33.2	28.3	71.1	0.025	12.50	rain,snow	
1	33.2	38.4	38.9	87.1	0.095	29.17	rain	
2	37.7	41.6	42.6	91.0	0.793	70.83	rain	
3	8.0	23.6	23.4	67.4	0.052	29.17	rain,snow	
4	7.0	11.2	11.5	68.4	0.056	25.00	snow	

...	...	...	...	...	...	...	...	...
2673	53.0	65.3	50.3	60.5	0.090	25.00	rain	
2674	63.2	72.6	55.3	55.6	0.000	0.00	rain	
2675	62.6	73.9	54.6	52.0	0.000	0.00	rain	
2676	61.1	65.9	56.1	71.0	0.014	4.17	rain	
2677	53.7	67.5	48.6	55.4	0.000	0.00	none	

	snow	snowdepth	cloudcover	uvindex	severerisk	moonphase	\
0	0.0	0.2	41.7	4	0.0	0.12	
1	0.0	0.0	87.9	1	0.0	0.16	
2	0.0	0.0	78.7	0	0.0	0.19	
3	0.0	0.0	73.0	4	0.0	0.23	
4	0.2	0.0	83.5	3	0.0	0.25	

...	...	...	...	...	...	...	...	...
2673	0.0	0.0	65.9	7	10.0	0.63		
2674	0.0	0.0	47.9	10	10.0	0.67		
2675	0.0	0.0	27.2	9	10.0	0.70		
2676	0.0	0.0	68.1	6	10.0	0.74		
2677	0.0	0.0	16.2	8	10.0	0.75		

	icon
0	rain
1	rain
2	rain
3	rain
4	snow

...	...
2673	rain
2674	partly-cloudy
2675	partly-cloudy
2676	rain
2677	clear

[2678 rows x 20 columns]

```
[ ]: overdose_df = pd.read_csv('/Users/hunterberberich/Desktop/Data/OverdoseDataPA1.
↳csv')
overdose_df
```

```
/var/folders/kp/5dkz8bcx6vx8fhrjwhg0gw0w0000gn/T/ipykernel_29043/1390839308.py:1
: DtypeWarning: Columns
(0,1,6,14,17,25,26,27,28,29,30,31,33,34,35,36,37,38,40,41) have mixed types.
Specify dtype option on import or set low_memory=False.
overdose_df =
pd.read_csv('/Users/hunterberberich/Desktop/Data/OverdoseDataPA1.csv')
```

```
[ ]: hbkk-dwy3_version_307  Unnamed: 1  Unnamed: 2  Unnamed: 3  \
0      dose_count  incident_id  incident_time      day
1              1          184      13:52:00  Wednesday
2              2         39237      4:13:00   Saturday
3              1         27008      0:15:00   Monday
4              2         47495     21:00:00   Saturday
...
53933          2         39326     14:27:00   Tuesday
53934          2         14560     16:36:00   Friday
53935          1         12293     21:30:00   Monday
53936          2         35341     21:58:00   Sunday
53937          0         25135     22:00:00   Friday

      Unnamed: 4  Unnamed: 5  Unnamed: 6  Unnamed: 7  \
0  incident_county_name  incident_state  victim_id  gender_desc
1      Allegheny  Pennsylvania      124      Female
2      Allegheny  Pennsylvania     31680      Female
3      Lackawanna  Pennsylvania     21028      Female
4      Adams  Pennsylvania     38590      Male
...
53933      Cambria  Pennsylvania     31751      Male
53934      York  Pennsylvania     11654      Male
53935      Allegheny  Pennsylvania     9750      Female
53936      Westmoreland  Pennsylvania     28171      Male
53937      Fayette  Pennsylvania     19712      Male

      Unnamed: 8  Unnamed: 9  ...  Unnamed: 32  \
0      age_range      race  ...  geocoded_column
1      20 - 24      WHITE  ...  POINT (-79.986198 40.467355)
2      50 - 59      WHITE  ...  POINT (-79.986198 40.467355)
3      30 - 39      BLACK  ...  POINT (-75.612183 41.439101)
4      40 - 49      WHITE  ...  POINT (-77.222243 39.872096)
...
53933      40 - 49      BLACK  ...  POINT (-78.718942 40.491275)
53934      25 - 29      WHITE  ...  POINT (-76.725761 39.921925)
53935      30 - 39      WHITE  ...  POINT (-79.986198 40.467355)
```

```

53936    40 - 49    WHITE ... POINT (-79.471341 40.310315)
53937    40 - 49    WHITE ... POINT (-79.651896 39.919448)

```

```

      Unnamed: 33      Unnamed: 34      Unnamed: 35  \
0      geocoded_column_address geocoded_column_city geocoded_column_zip
1              NaN              NaN              NaN
2              NaN              NaN              NaN
3              NaN              NaN              NaN
4              NaN              NaN              NaN
...
53933              NaN              NaN              NaN
53934              NaN              NaN              NaN
53935              NaN              NaN              NaN
53936              NaN              NaN              NaN
53937              NaN              NaN              NaN

```

```

      Unnamed: 36      Unnamed: 37      Unnamed: 38  \
0      geocoded_column_1_city geocoded_column_1_zip geocoded_column_1_state
1              NaN              NaN              NaN
2              NaN              NaN              NaN
3              NaN              NaN              NaN
4              NaN              NaN              NaN
...
53933              NaN              NaN              NaN
53934              NaN              NaN              NaN
53935              NaN              NaN              NaN
53936              NaN              NaN              NaN
53937              NaN              NaN              NaN

```

```

      Unnamed: 39 Unnamed: 40      Unnamed: 41
0      geocoded_column_1 dose_unit geocoded_column_1_address
1  POINT (-79.986198 40.467355)      0      NaN
2  POINT (-79.986198 40.467355)      0      NaN
3  POINT (-75.032709 41.332572)      0      NaN
4      POINT (0 0)      2      NaN
...
53933  POINT (-78.718942 40.491275)      2      NaN
53934  POINT (-76.725761 39.921925)      4      NaN
53935  POINT (-79.986198 40.467355)      0      NaN
53936  POINT (-79.471341 40.310315)      2      NaN
53937  POINT (-79.651896 39.919448)      0      NaN

```

```
[53938 rows x 42 columns]
```

```

[ ]: overdose_df.columns = overdose_df.iloc[0]
overdose_df = overdose_df[1:]
overdose_df.reset_index(drop=True, inplace=True)

```

```
[ ]: overdose_df
```

```
[ ]: 0      dose_count incident_id incident_time      day incident_county_name \
0          1          184      13:52:00  Wednesday      Allegheny
1          2         39237       4:13:00    Saturday      Allegheny
2          1         27008       0:15:00     Monday      Lackawanna
3          2         47495      21:00:00    Saturday        Adams
4          1         25677      17:30:00  Wednesday      Allegheny
...
53932      2         39326      14:27:00    Tuesday      Cambria
53933      2         14560      16:36:00    Friday        York
53934      1         12293      21:30:00    Monday      Allegheny
53935      2         35341      21:58:00    Sunday      Westmoreland
53936      0         25135      22:00:00    Friday      Fayette
```

```
0      incident_state victim_id gender_desc age_range  race ... \
0      Pennsylvania      124      Female  20 - 24  WHITE ...
1      Pennsylvania     31680      Female  50 - 59  WHITE ...
2      Pennsylvania     21028      Female  30 - 39  BLACK ...
3      Pennsylvania     38590       Male  40 - 49  WHITE ...
4      Pennsylvania     20086       Male  50 - 59  WHITE ...
...
53932  Pennsylvania     31751       Male  40 - 49  BLACK ...
53933  Pennsylvania     11654       Male  25 - 29  WHITE ...
53934  Pennsylvania      9750      Female  30 - 39  WHITE ...
53935  Pennsylvania     28171       Male  40 - 49  WHITE ...
53936  Pennsylvania     19712       Male  40 - 49  WHITE ...
```

```
0      geocoded_column geocoded_column_address \
0      POINT (-79.986198 40.467355)      NaN
1      POINT (-79.986198 40.467355)      NaN
2      POINT (-75.612183 41.439101)      NaN
3      POINT (-77.222243 39.872096)      NaN
4      POINT (-79.986198 40.467355)      NaN
...
53932  POINT (-78.718942 40.491275)      NaN
53933  POINT (-76.725761 39.921925)      NaN
53934  POINT (-79.986198 40.467355)      NaN
53935  POINT (-79.471341 40.310315)      NaN
53936  POINT (-79.651896 39.919448)      NaN
```

```
0      geocoded_column_city geocoded_column_zip geocoded_column_1_city \
0      NaN      NaN      NaN
1      NaN      NaN      NaN
2      NaN      NaN      NaN
3      NaN      NaN      NaN
4      NaN      NaN      NaN
```

```

...
53932      NaN      NaN      NaN
53933      NaN      NaN      NaN
53934      NaN      NaN      NaN
53935      NaN      NaN      NaN
53936      NaN      NaN      NaN

```

```

0      geocoded_column_1_zip geocoded_column_1_state \
0      NaN      NaN
1      NaN      NaN
2      NaN      NaN
3      NaN      NaN
4      NaN      NaN

```

```

...
53932      NaN      NaN
53933      NaN      NaN
53934      NaN      NaN
53935      NaN      NaN
53936      NaN      NaN

```

```

0      geocoded_column_1 dose_unit geocoded_column_1_address
0      POINT (-79.986198 40.467355)      0      NaN
1      POINT (-79.986198 40.467355)      0      NaN
2      POINT (-75.032709 41.332572)      0      NaN
3      POINT (0 0)      2      NaN
4      POINT (-79.986198 40.467355)      0      NaN
...
53932      POINT (-78.718942 40.491275)      2      NaN
53933      POINT (-76.725761 39.921925)      4      NaN
53934      POINT (-79.986198 40.467355)      0      NaN
53935      POINT (-79.471341 40.310315)      2      NaN
53936      POINT (-79.651896 39.919448)      0      NaN

```

[53937 rows x 42 columns]

```
[ ]: overdose_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 53937 entries, 0 to 53936
```

```
Data columns (total 42 columns):
```

#	Column	Non-Null Count	Dtype
0	dose_count	53937 non-null	object
1	incident_id	53937 non-null	object
2	incident_time	53937 non-null	object
3	day	53937 non-null	object
4	incident_county_name	53937 non-null	object
5	incident_state	53937 non-null	object

```

6   victim_id          53937 non-null object
7   gender_desc        53937 non-null object
8   age_range          53937 non-null object
9   race               53937 non-null object
10  ethnicity_desc     53937 non-null object
11  victim_state       48246 non-null object
12  victim_county      48246 non-null object
13  accidental_exposure 53937 non-null object
14  victim_od_drug_id  53937 non-null object
15  susp_od_drug_desc  53937 non-null object
16  naloxone_administered 53937 non-null object
17  administration_id  53937 non-null object
18  incident_date      53937 non-null object
19  dose_desc          35252 non-null object
20  response_time_desc 32434 non-null object
21  survive            53937 non-null object
22  response_desc      35274 non-null object
23  revive_action_desc 32434 non-null object
24  third_party_admin_desc 15230 non-null object
25  incident_county_fips_code 53937 non-null object
26  incident_county_lat 53937 non-null object
27  incident_county_long 53937 non-null object
28  victim_county_fips_code 48246 non-null object
29  victim_county_lat  53937 non-null object
30  victim_county_long 53937 non-null object
31  geocoded_column_state 0 non-null object
32  geocoded_column    53937 non-null object
33  geocoded_column_address 0 non-null object
34  geocoded_column_city 0 non-null object
35  geocoded_column_zip 0 non-null object
36  geocoded_column_1_city 0 non-null object
37  geocoded_column_1_zip 0 non-null object
38  geocoded_column_1_state 0 non-null object
39  geocoded_column_1   53937 non-null object
40  dose_unit           53937 non-null object
41  geocoded_column_1_address 0 non-null object
dtypes: object(42)
memory usage: 17.3+ MB

```

```
[ ]: overdose_df2 = overdose_df.drop(overdose_df.columns[31:40], axis=1)
```

```
[ ]: overdose_df2
```

```
[ ]: 0      dose_count incident_id incident_time      day incident_county_name \
0           1         184      13:52:00 Wednesday      Allegheny
1           2        39237       4:13:00 Saturday      Allegheny
2           1        27008       0:15:00 Monday      Lackawanna
```

3	2	47495	21:00:00	Saturday	Adams
4	1	25677	17:30:00	Wednesday	Allegheny
...	...	...	...	...	...
53932	2	39326	14:27:00	Tuesday	Cambria
53933	2	14560	16:36:00	Friday	York
53934	1	12293	21:30:00	Monday	Allegheny
53935	2	35341	21:58:00	Sunday	Westmoreland
53936	0	25135	22:00:00	Friday	Fayette

0	incident_state	victim_id	gender_desc	age_range	race	...	\
0	Pennsylvania	124	Female	20 - 24	WHITE	...	
1	Pennsylvania	31680	Female	50 - 59	WHITE	...	
2	Pennsylvania	21028	Female	30 - 39	BLACK	...	
3	Pennsylvania	38590	Male	40 - 49	WHITE	...	
4	Pennsylvania	20086	Male	50 - 59	WHITE	...	
...	...	...	...	...	...	...	
53932	Pennsylvania	31751	Male	40 - 49	BLACK	...	
53933	Pennsylvania	11654	Male	25 - 29	WHITE	...	
53934	Pennsylvania	9750	Female	30 - 39	WHITE	...	
53935	Pennsylvania	28171	Male	40 - 49	WHITE	...	
53936	Pennsylvania	19712	Male	40 - 49	WHITE	...	

0	revive_action_desc	third_party_admin_desc	\
0	NaN	NaN	
1	HOSPITAL CONSCIOUS	GOOD SAMARITAN	
2	REFUSED TRANSPORT	GOOD SAMARITAN	
3	TRANSPORTED TO TREATMENT	EMS	
4	HOSPITAL UNCONSCIOUS	FIRE DEPARTMENT	
...	...	...	
53932	NaN	NaN	
53933	REFUSED TRANSPORT	NaN	
53934	HOSPITAL CONSCIOUS	GOOD SAMARITAN	
53935	HOSPITAL CONSCIOUS	EMS	
53936	NaN	NaN	

0	incident_county_fips_code	incident_county_lat	incident_county_long	\
0	3	40.467355	-79.986198	
1	3	40.467355	-79.986198	
2	69	41.439101	-75.612183	
3	1	39.872096	-77.222243	
4	3	40.467355	-79.986198	
...	...	...	...	
53932	21	40.491275	-78.718942	
53933	133	39.921925	-76.725761	
53934	3	40.467355	-79.986198	
53935	129	40.310315	-79.471341	
53936	51	39.919448	-79.651896	

	victim_county_fips_code	victim_county_lat	victim_county_long	dose_unit	\
0	3	40.467355	-79.986198	0	
1	3	40.467355	-79.986198	0	
2	103	41.332572	-75.032709	0	
3	NaN	0	0	2	
4	3	40.467355	-79.986198	0	
...	...	...	...	...	
53932	21.0	40.491275	-78.718942	2	
53933	133.0	39.921925	-76.725761	4	
53934	3.0	40.467355	-79.986198	0	
53935	129.0	40.310315	-79.471341	2	
53936	51.0	39.919448	-79.651896	0	

	geocoded_column_1_address
0	NaN
1	NaN
2	NaN
3	NaN
4	NaN
...	...
53932	NaN
53933	NaN
53934	NaN
53935	NaN
53936	NaN

[53937 rows x 33 columns]

[]:

[]: overdose_df2.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 53937 entries, 0 to 53936
Data columns (total 32 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   dose_count                            53937 non-null  object
1   incident_id                           53937 non-null  object
2   incident_time                          53937 non-null  object
3   day                                    53937 non-null  object
4   incident_county_name                  53937 non-null  object
5   incident_state                         53937 non-null  object
6   victim_id                             53937 non-null  object
7   gender_desc                           53937 non-null  object
8   age_range                             53937 non-null  object
9   race                                  53937 non-null  object
```

10	ethnicity_desc	53937	non-null	object
11	victim_state	48246	non-null	object
12	victim_county	48246	non-null	object
13	accidental_exposure	53937	non-null	object
14	victim_od_drug_id	53937	non-null	object
15	susp_od_drug_desc	53937	non-null	object
16	naloxone_administered	53937	non-null	object
17	administration_id	53937	non-null	object
18	dose_desc	35252	non-null	object
19	response_time_desc	32434	non-null	object
20	survive	53937	non-null	object
21	response_desc	35274	non-null	object
22	revive_action_desc	32434	non-null	object
23	third_party_admin_desc	15230	non-null	object
24	incident_county_fips_code	53937	non-null	object
25	incident_county_lat	53937	non-null	object
26	incident_county_long	53937	non-null	object
27	victim_county_fips_code	48246	non-null	object
28	victim_county_lat	53937	non-null	object
29	victim_county_long	53937	non-null	object
30	dose_unit	53937	non-null	object
31	geocoded_column_1_address	0	non-null	object

dtypes: object(32)

memory usage: 13.2+ MB

```
[ ]: overdose_df4 = overdose_df.drop(overdose_df.columns[31:40], axis=1)
```

```
[ ]: overdose_df4.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 53937 entries, 0 to 53936
```

```
Data columns (total 33 columns):
```

#	Column	Non-Null Count	Dtype
0	dose_count	53937 non-null	object
1	incident_id	53937 non-null	object
2	incident_time	53937 non-null	object
3	day	53937 non-null	object
4	incident_county_name	53937 non-null	object
5	incident_state	53937 non-null	object
6	victim_id	53937 non-null	object
7	gender_desc	53937 non-null	object
8	age_range	53937 non-null	object
9	race	53937 non-null	object
10	ethnicity_desc	53937 non-null	object
11	victim_state	48246 non-null	object
12	victim_county	48246 non-null	object
13	accidental_exposure	53937 non-null	object

```

14 victim_od_drug_id      53937 non-null object
15 susp_od_drug_desc     53937 non-null object
16 naloxone_administered 53937 non-null object
17 administration_id     53937 non-null object
18 incident_date          53937 non-null object
19 dose_desc              35252 non-null object
20 response_time_desc     32434 non-null object
21 survive                53937 non-null object
22 response_desc          35274 non-null object
23 revive_action_desc     32434 non-null object
24 third_party_admin_desc 15230 non-null object
25 incident_county_fips_code 53937 non-null object
26 incident_county_lat    53937 non-null object
27 incident_county_long   53937 non-null object
28 victim_county_fips_code 48246 non-null object
29 victim_county_lat      53937 non-null object
30 victim_county_long     53937 non-null object
31 dose_unit              53937 non-null object
32 geocoded_column_1_address 0 non-null      object
dtypes: object(33)
memory usage: 13.6+ MB

```

```
[ ]: overdose_df5 = overdose_df4.drop(overdose_df4.columns[25:31], axis=1)
overdose_df5
```

```
[ ]: 0      dose_count incident_id incident_time      day incident_county_name \
0           1         184      13:52:00  Wednesday      Allegheny
1           2        39237       4:13:00   Saturday      Allegheny
2           1        27008       0:15:00    Monday      Lackawanna
3           2        47495      21:00:00   Saturday          Adams
4           1        25677      17:30:00   Wednesday      Allegheny
...      ...      ...      ...      ...      ...
53932        2        39326      14:27:00    Tuesday      Cambria
53933        2        14560      16:36:00    Friday      York
53934        1        12293      21:30:00    Monday      Allegheny
53935        2        35341      21:58:00    Sunday      Westmoreland
53936        0        25135      22:00:00    Friday      Fayette

0      incident_state victim_id gender_desc age_range  race ... \
0      Pennsylvania     124      Female    20 - 24  WHITE ...
1      Pennsylvania    31680      Female    50 - 59  WHITE ...
2      Pennsylvania    21028      Female    30 - 39  BLACK ...
3      Pennsylvania    38590       Male    40 - 49  WHITE ...
4      Pennsylvania    20086       Male    50 - 59  WHITE ...
...      ...      ...      ...      ...      ...
53932  Pennsylvania    31751       Male    40 - 49  BLACK ...
53933  Pennsylvania    11654       Male    25 - 29  WHITE ...

```

53934	Pennsylvania	9750	Female	30 - 39	WHITE	...
53935	Pennsylvania	28171	Male	40 - 49	WHITE	...
53936	Pennsylvania	19712	Male	40 - 49	WHITE	...

	administration_id	incident_date	dose_desc	response_time_desc	survive	\
0	26096	3/7/2018	UNKNOWN	NaN	N	
1	26073	4/22/2023	UNKNOWN	1-3 MINUTES	U	
2	26027	8/2/2021	UNKNOWN	1-3 MINUTES	Y	
3	26115	3/23/2024	MG	3-5 MINUTES	Y	
4	26029	5/26/2021	UNKNOWN	1-3 MINUTES	U	
...	...	...	...	...	...	
53932	21291	4/25/2023	MG	NaN	N	
53933	7915	1/24/2020	MG	1-3 MINUTES	Y	
53934	6642	9/30/2019	UNKNOWN	DON'T KNOW	Y	
53935	19125	11/27/2022	MG	>5 MINUTES	Y	
53936	0	4/2/2021	NaN	NaN	Y	

	response_desc	revive_action_desc	\
0	NO RESPONSE TO NALOXONE	NaN	
1	RESPONSIVE BUT SEDATED	HOSPITAL CONSCIOUS	
2	RESPONSIVE AND ALERT	REFUSED TRANSPORT	
3	RESPONSIVE BUT SEDATED	TRANSPORTED TO TREATMENT	
4	UNKNOWN	HOSPITAL UNCONSCIOUS	
...	...	...	
53932	NO RESPONSE TO NALOXONE	NaN	
53933	COMBATIVE	REFUSED TRANSPORT	
53934	RESPONSIVE AND ALERT	HOSPITAL CONSCIOUS	
53935	RESPONSIVE AND ALERT	HOSPITAL CONSCIOUS	
53936	NaN	NaN	

	third_party_admin_desc	dose_unit	geocoded_column_1_address
0	NaN	0	NaN
1	GOOD SAMARITAN	0	NaN
2	GOOD SAMARITAN	0	NaN
3	EMS	2	NaN
4	FIRE DEPARTMENT	0	NaN
...	...	...	...
53932	NaN	2	NaN
53933	NaN	4	NaN
53934	GOOD SAMARITAN	0	NaN
53935	EMS	2	NaN
53936	NaN	0	NaN

[53937 rows x 27 columns]

```
[ ]: overdose_df3 = overdose_df3.drop(overdose_df3.columns[28], axis=1)
```

```
[ ]: overdose_df3
```

```
[ ]: Empty DataFrame
```

```
Columns: []
```

```
Index: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, ...]
```

```
[53937 rows x 0 columns]
```

```
[ ]: overdose_df6 = overdose_df5.drop(overdose_df5.columns[-1], axis=1)
overdose_df6
```

```
[ ]: 0      dose_count incident_id incident_time      day incident_county_name \
0              1         184      13:52:00  Wednesday      Allegheny
1              2        39237       4:13:00   Saturday      Allegheny
2              1        27008       0:15:00    Monday    Lackawanna
3              2        47495      21:00:00   Saturday        Adams
4              1        25677      17:30:00   Wednesday      Allegheny
```

```
...      ...      ...      ...      ...      ...
53932          2        39326      14:27:00    Tuesday      Cambria
53933          2        14560      16:36:00    Friday        York
53934          1        12293      21:30:00    Monday      Allegheny
53935          2        35341      21:58:00    Sunday    Westmoreland
53936          0        25135      22:00:00    Friday      Fayette
```

```
0      incident_state victim_id gender_desc age_range  race  ... \
0      Pennsylvania      124      Female  20 - 24  WHITE  ...
1      Pennsylvania    31680      Female  50 - 59  WHITE  ...
2      Pennsylvania    21028      Female  30 - 39  BLACK  ...
3      Pennsylvania    38590       Male  40 - 49  WHITE  ...
4      Pennsylvania    20086       Male  50 - 59  WHITE  ...
```

```
...      ...      ...      ...      ...      ...
53932  Pennsylvania    31751       Male  40 - 49  BLACK  ...
53933  Pennsylvania    11654       Male  25 - 29  WHITE  ...
53934  Pennsylvania     9750      Female  30 - 39  WHITE  ...
53935  Pennsylvania    28171       Male  40 - 49  WHITE  ...
53936  Pennsylvania    19712       Male  40 - 49  WHITE  ...
```

```
0      naloxone_administered administration_id incident_date dose_desc \
0              Y              26096      3/7/2018    UNKNOWN
1              Y              26073      4/22/2023    UNKNOWN
2              Y              26027      8/2/2021    UNKNOWN
3              Y              26115      3/23/2024        MG
```

4		Y	26029	5/26/2021	UNKNOWN
...	...		...	...	
53932		Y	21291	4/25/2023	MG
53933		Y	7915	1/24/2020	MG
53934		Y	6642	9/30/2019	UNKNOWN
53935		Y	19125	11/27/2022	MG
53936		N	0	4/2/2021	NaN

	response_time_desc	survive	response_desc	\
0	NaN	N	NO RESPONSE TO NALOXONE	
1	1-3 MINUTES	U	RESPONSIVE BUT SEDATED	
2	1-3 MINUTES	Y	RESPONSIVE AND ALERT	
3	3-5 MINUTES	Y	RESPONSIVE BUT SEDATED	
4	1-3 MINUTES	U	UNKNOWN	
...	...	...	...	
53932	NaN	N	NO RESPONSE TO NALOXONE	
53933	1-3 MINUTES	Y	COMBATIVE	
53934	DON'T KNOW	Y	RESPONSIVE AND ALERT	
53935	>5 MINUTES	Y	RESPONSIVE AND ALERT	
53936	NaN	Y	NaN	

	revive_action_desc	third_party_admin_desc	dose_unit
0	NaN	NaN	0
1	HOSPITAL CONSCIOUS	GOOD SAMARITAN	0
2	REFUSED TRANSPORT	GOOD SAMARITAN	0
3	TRANSPORTED TO TREATMENT	EMS	2
4	HOSPITAL UNCONSCIOUS	FIRE DEPARTMENT	0
...	...	...	...
53932	NaN	NaN	2
53933	REFUSED TRANSPORT	NaN	4
53934	HOSPITAL CONSCIOUS	GOOD SAMARITAN	0
53935	HOSPITAL CONSCIOUS	EMS	2
53936	NaN	NaN	0

[53937 rows x 26 columns]

```
[ ]: overdose_df6.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 53937 entries, 0 to 53936
```

```
Data columns (total 26 columns):
```

#	Column	Non-Null Count	Dtype
---	-----	-----	-----
0	dose_count	53937 non-null	object
1	incident_id	53937 non-null	object
2	incident_time	53937 non-null	object
3	day	53937 non-null	object
4	incident_county_name	53937 non-null	object

```

5  incident_state      53937 non-null object
6  victim_id           53937 non-null object
7  gender_desc         53937 non-null object
8  age_range           53937 non-null object
9  race                53937 non-null object
10 ethnicity_desc      53937 non-null object
11 victim_state        48246 non-null object
12 victim_county       48246 non-null object
13 accidental_exposure 53937 non-null object
14 victim_od_drug_id   53937 non-null object
15 susp_od_drug_desc   53937 non-null object
16 naloxone_administered 53937 non-null object
17 administration_id   53937 non-null object
18 incident_date        53937 non-null object
19 dose_desc            35252 non-null object
20 response_time_desc   32434 non-null object
21 survive              53937 non-null object
22 response_desc        35274 non-null object
23 revive_action_desc   32434 non-null object
24 third_party_admin_desc 15230 non-null object
25 dose_unit            53937 non-null object

```

dtypes: object(26)

memory usage: 10.7+ MB

```

[ ]: columns_to_drop2 = ['response_desc', 'revive_action_desc',
    ↳ 'third_party_admin_desc', 'victim_state', 'victim_county',
    ↳ 'dose_desc', 'response_time_desc']
overdose_df7 = overdose_df6.drop(columns=columns_to_drop2)
overdose_df7

```

```

[ ]: 0  dose_count incident_id incident_time      day incident_county_name \
0          1         184      13:52:00  Wednesday      Allegheny
1          2        39237       4:13:00   Saturday      Allegheny
2          1        27008       0:15:00    Monday      Lackawanna
3          2        47495      21:00:00   Saturday          Adams
4          1        25677      17:30:00   Wednesday      Allegheny
...
53932      2        39326      14:27:00    Tuesday      Cambria
53933      2        14560      16:36:00    Friday          York
53934      1        12293      21:30:00    Monday      Allegheny
53935      2        35341      21:58:00    Sunday  Westmoreland
53936      0        25135      22:00:00    Friday      Fayette

0  incident_state victim_id gender_desc age_range  race \
0  Pennsylvania    124      Female  20 - 24  WHITE
1  Pennsylvania   31680      Female  50 - 59  WHITE
2  Pennsylvania   21028      Female  30 - 39  BLACK

```

3	Pennsylvania	38590	Male	40 - 49	WHITE
4	Pennsylvania	20086	Male	50 - 59	WHITE
...	...	...	...	...	...
53932	Pennsylvania	31751	Male	40 - 49	BLACK
53933	Pennsylvania	11654	Male	25 - 29	WHITE
53934	Pennsylvania	9750	Female	30 - 39	WHITE
53935	Pennsylvania	28171	Male	40 - 49	WHITE
53936	Pennsylvania	19712	Male	40 - 49	WHITE

0	ethnicity_desc	accidental_exposure	victim_od_drug_id	\
0	Unknown		N	155
1	Unknown		N	45381
2	NON-HISPANIC OR NOT LATINO		N	29279
3	NON-HISPANIC OR NOT LATINO		N	55871
4	NON-HISPANIC OR NOT LATINO		N	27836
...	...		...	...
53932	NON-HISPANIC OR NOT LATINO		N	45525
53933	NON-HISPANIC OR NOT LATINO		N	15624
53934	NON-HISPANIC OR NOT LATINO		N	12964
53935	NON-HISPANIC OR NOT LATINO		N	39986
53936	NON-HISPANIC OR NOT LATINO		N	27300

0	susp_od_drug_desc	naloxone_administered	administration_id	incident_date	\
0	HEROIN	Y	26096	3/7/2018	
1	HEROIN	Y	26073	4/22/2023	
2	HEROIN	Y	26027	8/2/2021	
3	UNKNOWN	Y	26115	3/23/2024	
4	HEROIN	Y	26029	5/26/2021	
...	...	...	...	...	
53932	HEROIN	Y	21291	4/25/2023	
53933	FENTANYL	Y	7915	1/24/2020	
53934	HEROIN	Y	6642	9/30/2019	
53935	HEROIN	Y	19125	11/27/2022	
53936	HEROIN	N	0	4/2/2021	

0	survive	dose_unit
0	N	0
1	U	0
2	Y	0
3	Y	2
4	U	0
...	...	...
53932	N	2
53933	Y	4
53934	Y	0
53935	Y	2
53936	Y	0

[53937 rows x 19 columns]

```
[ ]: overdose_df7['incident_date'] = pd.to_datetime(overdose_df7['incident_date'],  
      ↪errors='coerce')
```

```
[ ]: incident_date_column = overdose_df7.pop('incident_date')  
      overdose_df7.insert(0, 'incident_date', incident_date_column)  
      overdose_df7
```

```
[ ]: 0      incident_date dose_count incident_id incident_time      day \  
      0      2018-03-07          1         184      13:52:00  Wednesday  
      1      2023-04-22          2        39237       4:13:00   Saturday  
      2      2021-08-02          1        27008       0:15:00    Monday  
      3      2024-03-23          2        47495      21:00:00   Saturday  
      4      2021-05-26          1        25677      17:30:00   Wednesday  
      ...      ...      ...      ...      ...      ...  
      53932      2023-04-25          2        39326      14:27:00   Tuesday  
      53933      2020-01-24          2        14560      16:36:00    Friday  
      53934      2019-09-30          1        12293      21:30:00    Monday  
      53935      2022-11-27          2        35341      21:58:00    Sunday  
      53936      2021-04-02          0        25135      22:00:00    Friday  
  
      0      incident_county_name incident_state victim_id gender_desc age_range \  
      0      Allegheny      Pennsylvania      124      Female      20 - 24  
      1      Allegheny      Pennsylvania      31680     Female      50 - 59  
      2      Lackawanna      Pennsylvania      21028     Female      30 - 39  
      3      Adams      Pennsylvania      38590      Male      40 - 49  
      4      Allegheny      Pennsylvania      20086      Male      50 - 59  
      ...      ...      ...      ...      ...      ...  
      53932      Cambria      Pennsylvania      31751      Male      40 - 49  
      53933      York      Pennsylvania      11654      Male      25 - 29  
      53934      Allegheny      Pennsylvania      9750      Female      30 - 39  
      53935      Westmoreland      Pennsylvania      28171      Male      40 - 49  
      53936      Fayette      Pennsylvania      19712      Male      40 - 49  
  
      0      race      ethnicity_desc accidental_exposure \  
      0      WHITE      Unknown      N  
      1      WHITE      Unknown      N  
      2      BLACK      NON-HISPANIC OR NOT LATINO      N  
      3      WHITE      NON-HISPANIC OR NOT LATINO      N  
      4      WHITE      NON-HISPANIC OR NOT LATINO      N  
      ...      ...      ...      ...  
      53932      BLACK      NON-HISPANIC OR NOT LATINO      N  
      53933      WHITE      NON-HISPANIC OR NOT LATINO      N  
      53934      WHITE      NON-HISPANIC OR NOT LATINO      N  
      53935      WHITE      NON-HISPANIC OR NOT LATINO      N
```

53936 WHITE NON-HISPANIC OR NOT LATINO N

	victim_od_drug_id	susp_od_drug_desc	naloxone_administered	\
0	155	HEROIN	Y	
1	45381	HEROIN	Y	
2	29279	HEROIN	Y	
3	55871	UNKNOWN	Y	
4	27836	HEROIN	Y	
...	...	...	...	
53932	45525	HEROIN	Y	
53933	15624	FENTANYL	Y	
53934	12964	HEROIN	Y	
53935	39986	HEROIN	Y	
53936	27300	HEROIN	N	

	administration_id	survive	dose_unit
0	26096	N	0
1	26073	U	0
2	26027	Y	0
3	26115	Y	2
4	26029	U	0
...	...	...	...
53932	21291	N	2
53933	7915	Y	4
53934	6642	Y	0
53935	19125	Y	2
53936	0	Y	0

[53937 rows x 19 columns]

```
[ ]: overdose_df7['race'].unique()
```

```
[ ]: array(['WHITE', 'BLACK', 'UNKNOWN', 'ASIAN', 'AMERICAN INDIAN'],
dtype=object)
```

```
[ ]: df_alleggheny = overdose_df7[overdose_df7['incident_county_name'] == 'Allegheny']
df_alleggheny
```

	incident_date	dose_count	incident_id	incident_time	day	\
0	2018-03-07	1	184	13:52:00	Wednesday	
1	2023-04-22	2	39237	4:13:00	Saturday	
4	2021-05-26	1	25677	17:30:00	Wednesday	
6	2018-08-03	1	3984	0:02:00	Friday	
22	2023-09-04	0	42561	8:00:00	Monday	
...	...	...	...	...	...	
53897	2021-07-24	2	26867	0:00:00	Saturday	
53904	2023-05-08	2	39762	23:00:00	Monday	

53918	2021-02-20	0	23593	1:00:00	Saturday
53929	2023-04-16	1	39064	21:16:00	Sunday
53934	2019-09-30	1	12293	21:30:00	Monday

0	incident_county_name	incident_state	victim_id	gender_desc	age_range	\
0	Allegheny	Pennsylvania	124	Female	20 - 24	
1	Allegheny	Pennsylvania	31680	Female	50 - 59	
4	Allegheny	Pennsylvania	20086	Male	50 - 59	
6	Allegheny	Pennsylvania	3252	Male	50 - 59	
22	Allegheny	Pennsylvania	34747	Male	40 - 49	
...	...	...	...	...	...	
53897	Allegheny	Pennsylvania	20932	Female	20 - 24	
53904	Allegheny	Pennsylvania	32155	Male	30 - 39	
53918	Allegheny	Pennsylvania	18642	Male	25 - 29	
53929	Allegheny	Pennsylvania	31523	Male	40 - 49	
53934	Allegheny	Pennsylvania	9750	Female	30 - 39	

0	race	ethnicity_desc	accidental_exposure	\
0	WHITE	Unknown	N	
1	WHITE	Unknown	N	
4	WHITE	NON-HISPANIC OR NOT LATINO	N	
6	WHITE	Unknown	N	
22	WHITE	NON-HISPANIC OR NOT LATINO	N	
...	...	...	...	
53897	WHITE	NON-HISPANIC OR NOT LATINO	N	
53904	WHITE	NON-HISPANIC OR NOT LATINO	N	
53918	WHITE	NON-HISPANIC OR NOT LATINO	N	
53929	WHITE	NON-HISPANIC OR NOT LATINO	N	
53934	WHITE	NON-HISPANIC OR NOT LATINO	N	

0	victim_od_drug_id	susp_od_drug_desc	naloxone_administered	\
0	155	HEROIN	Y	
1	45381	HEROIN	Y	
4	27836	HEROIN	Y	
6	4195	HEROIN	Y	
22	50059	HEROIN	N	
...	...	...	...	
53897	29135	UNKNOWN	Y	
53904	46153	FENTANYL	Y	
53918	25615	ALCOHOL	N	
53929	45151	HEROIN	Y	
53934	12964	HEROIN	Y	

0	administration_id	survive	dose_unit
0	26096	N	0
1	26073	U	0
4	26029	U	0

6	26072	N	0
22	0	N	0
...	...	...	...
53897	14698	Y	4
53904	21537	Y	4
53918	0	Y	0
53929	21138	Y	4
53934	6642	Y	0

[7204 rows x 19 columns]

```
[ ]: df_alleggheny['susp_od_drug_desc'].unique()
```

```
[ ]: array(['HEROIN', 'FENTANYL', 'PHARMACEUTICAL OTHER', 'UNKNOWN',
          'MARIJUANA', 'CARFENTANIL',
          'FENTANYL ANALOG/OTHER SYNTHETIC OPIOID', 'METHADONE', 'ALCOHOL',
          'OTHER', 'BENZODIAZEPINES (I.E. VALIUM, XANAX, ATIVAN, ETC)',
          'COCAINE/CRACK', 'PHARMACEUTICAL OPIOID', 'METHAMPHETAMINE',
          'SYNTHETIC MARIJUANA', 'PHARMACEUTICAL STIMULANT',
          'BARBITURATES (I.E. AMYTAL, NEMBUTAL, ETC)',
          'BUPRENORPHINE (I.E. SUBOXONE, SUBLOCADE, SUBUTEX, BUTRANS, BUPRENEX,
          ETC)',
          'XYLAZINE', 'BATH SALTS'], dtype=object)
```

```
[ ]: opioid_related = [
      'HEROIN', 'FENTANYL', 'CARFENTANIL',
      'FENTANYL ANALOG/OTHER SYNTHETIC OPIOID', 'METHADONE',
      'PHARMACEUTICAL OPIOID',
      'BUPRENORPHINE (I.E. SUBOXONE, SUBLOCADE, SUBUTEX, BUTRANS, BUPRENEX, ETC)']
```

```
[ ]: overdose_cleaned = df_alleggheny[df_alleggheny['susp_od_drug_desc'].
    ↪isin(opioid_related)]
overdose_cleaned
```

[]:	0	incident_date	dose_count	incident_id	incident_time	day	\
	0	2018-03-07	1	184	13:52:00	Wednesday	
	1	2023-04-22	2	39237	4:13:00	Saturday	
	4	2021-05-26	1	25677	17:30:00	Wednesday	
	6	2018-08-03	1	3984	0:02:00	Friday	
	22	2023-09-04	0	42561	8:00:00	Monday	
	...	...	...	...	...	...	
	53879	2023-06-20	0	40768	17:08:00	Tuesday	
	53891	2023-02-17	2	37405	16:00:00	Friday	
	53904	2023-05-08	2	39762	23:00:00	Monday	
	53929	2023-04-16	1	39064	21:16:00	Sunday	
	53934	2019-09-30	1	12293	21:30:00	Monday	

	incident_county_name	incident_state	victim_id	gender_desc	age_range	\
0	Allegheny	Pennsylvania	124	Female	20 - 24	
1	Allegheny	Pennsylvania	31680	Female	50 - 59	
4	Allegheny	Pennsylvania	20086	Male	50 - 59	
6	Allegheny	Pennsylvania	3252	Male	50 - 59	
22	Allegheny	Pennsylvania	34747	Male	40 - 49	
...	...	...	...	...	...	
53879	Allegheny	Pennsylvania	33077	Male	30 - 39	
53891	Allegheny	Pennsylvania	30037	Male	30 - 39	
53904	Allegheny	Pennsylvania	32155	Male	30 - 39	
53929	Allegheny	Pennsylvania	31523	Male	40 - 49	
53934	Allegheny	Pennsylvania	9750	Female	30 - 39	

	race	ethnicity_desc	accidental_exposure	\
0	WHITE	Unknown	N	
1	WHITE	Unknown	N	
4	WHITE	NON-HISPANIC OR NOT LATINO	N	
6	WHITE	Unknown	N	
22	WHITE	NON-HISPANIC OR NOT LATINO	N	
...	...	...	...	
53879	WHITE	NON-HISPANIC OR NOT LATINO	N	
53891	WHITE	NON-HISPANIC OR NOT LATINO	N	
53904	WHITE	NON-HISPANIC OR NOT LATINO	N	
53929	WHITE	NON-HISPANIC OR NOT LATINO	N	
53934	WHITE	NON-HISPANIC OR NOT LATINO	N	

	victim_od_drug_id	susp_od_drug_desc	naloxone_administered	\
0	155	HEROIN	Y	
1	45381	HEROIN	Y	
4	27836	HEROIN	Y	
6	4195	HEROIN	Y	
22	50059	HEROIN	N	
...	...	...	...	
53879	47582	HEROIN	N	
53891	42881	HEROIN	Y	
53904	46153	FENTANYL	Y	
53929	45151	HEROIN	Y	
53934	12964	HEROIN	Y	

	administration_id	survive	dose_unit
0	26096	N	0
1	26073	U	0
4	26029	U	0
6	26072	N	0
22	0	N	0
...	...	...	...
53879	0	Y	0

53891	20253	Y	4
53904	21537	Y	4
53929	21138	Y	4
53934	6642	Y	0

[5313 rows x 19 columns]

```
[ ]: overdose_cleaned = overdose_cleaned.rename(columns={'incident_date': 'date'})
overdose_cleaned
```

```
[ ]: 0      date dose_count incident_id incident_time      day \
0      2018-03-07          1          184      13:52:00  Wednesday
1      2023-04-22          2         39237       4:13:00   Saturday
4      2021-05-26          1         25677      17:30:00  Wednesday
6      2018-08-03          1          3984       0:02:00    Friday
22     2023-09-04          0         42561       8:00:00    Monday
...     ...           ...           ...           ...
53879  2023-06-20          0         40768      17:08:00   Tuesday
53891  2023-02-17          2         37405      16:00:00    Friday
53904  2023-05-08          2         39762      23:00:00    Monday
53929  2023-04-16          1         39064      21:16:00    Sunday
53934  2019-09-30          1         12293      21:30:00    Monday

0      incident_county_name incident_state victim_id gender_desc age_range \
0      Allegheny      Pennsylvania      124      Female      20 - 24
1      Allegheny      Pennsylvania     31680      Female      50 - 59
4      Allegheny      Pennsylvania     20086       Male      50 - 59
6      Allegheny      Pennsylvania      3252       Male      50 - 59
22     Allegheny      Pennsylvania     34747       Male      40 - 49
...     ...           ...           ...           ...
53879      Allegheny      Pennsylvania     33077       Male      30 - 39
53891      Allegheny      Pennsylvania     30037       Male      30 - 39
53904      Allegheny      Pennsylvania     32155       Male      30 - 39
53929      Allegheny      Pennsylvania     31523       Male      40 - 49
53934      Allegheny      Pennsylvania      9750      Female      30 - 39

0      race      ethnicity_desc accidental_exposure \
0      WHITE      Unknown      N
1      WHITE      Unknown      N
4      WHITE  NON-HISPANIC OR NOT LATINO      N
6      WHITE      Unknown      N
22     WHITE  NON-HISPANIC OR NOT LATINO      N
...     ...           ...           ...
53879  WHITE  NON-HISPANIC OR NOT LATINO      N
53891  WHITE  NON-HISPANIC OR NOT LATINO      N
53904  WHITE  NON-HISPANIC OR NOT LATINO      N
53929  WHITE  NON-HISPANIC OR NOT LATINO      N
```

53934 WHITE NON-HISPANIC OR NOT LATINO N

	victim_od_drug_id	susp_od_drug_desc	naloxone_administered	\
0	155	HEROIN	Y	
1	45381	HEROIN	Y	
4	27836	HEROIN	Y	
6	4195	HEROIN	Y	
22	50059	HEROIN	N	
...	...	...	...	
53879	47582	HEROIN	N	
53891	42881	HEROIN	Y	
53904	46153	FENTANYL	Y	
53929	45151	HEROIN	Y	
53934	12964	HEROIN	Y	

	administration_id	survive	dose_unit
0	26096	N	0
1	26073	U	0
4	26029	U	0
6	26072	N	0
22	0	N	0
...	...	...	...
53879	0	Y	0
53891	20253	Y	4
53904	21537	Y	4
53929	21138	Y	4
53934	6642	Y	0

[5313 rows x 19 columns]

```
[ ]: weather_df_allegheny2 = weather_df_allegheny2.rename(columns={'datetime':  
↳ 'date'})  
weather_df_allegheny2
```

	name	date	tempmax	tempmin	temp	feelslikemax	\
0	Allegheny County	2017-01-01	43.9	31.4	37.1	40.1	
1	Allegheny County	2017-01-02	46.3	37.0	42.6	43.3	
2	Allegheny County	2017-01-03	48.6	42.9	45.1	46.3	
3	Allegheny County	2017-01-04	44.3	21.1	33.8	37.4	
4	Allegheny County	2017-01-05	22.6	19.0	20.6	20.8	
...	...	...	...	...	...	...	
2673	Allegheny County	2024-04-27	79.1	53.0	65.3	79.1	
2674	Allegheny County	2024-04-28	82.0	63.2	72.6	81.5	
2675	Allegheny County	2024-04-29	82.9	62.6	74.1	82.1	
2676	Allegheny County	2024-04-30	72.3	61.1	65.9	72.3	
2677	Allegheny County	2024-05-01	79.9	53.7	67.5	79.9	

	feelslikemin	feelslike	dew	humidity	precip	precipcover	preciptype \
0	29.0	33.2	28.3	71.1	0.025	12.50	rain,snow
1	33.2	38.4	38.9	87.1	0.095	29.17	rain
2	37.7	41.6	42.6	91.0	0.793	70.83	rain
3	8.0	23.6	23.4	67.4	0.052	29.17	rain,snow
4	7.0	11.2	11.5	68.4	0.056	25.00	snow
...	...	...	...	...	...	...	
2673	53.0	65.3	50.3	60.5	0.090	25.00	rain
2674	63.2	72.6	55.3	55.6	0.000	0.00	rain
2675	62.6	73.9	54.6	52.0	0.000	0.00	rain
2676	61.1	65.9	56.1	71.0	0.014	4.17	rain
2677	53.7	67.5	48.6	55.4	0.000	0.00	none

	snow	snowdepth	cloudcover	uvindex	severerisk	moonphase \
0	0.0	0.2	41.7	4	0.0	0.12
1	0.0	0.0	87.9	1	0.0	0.16
2	0.0	0.0	78.7	0	0.0	0.19
3	0.0	0.0	73.0	4	0.0	0.23
4	0.2	0.0	83.5	3	0.0	0.25
...	...	...	...	...	...	
2673	0.0	0.0	65.9	7	10.0	0.63
2674	0.0	0.0	47.9	10	10.0	0.67
2675	0.0	0.0	27.2	9	10.0	0.70
2676	0.0	0.0	68.1	6	10.0	0.74
2677	0.0	0.0	16.2	8	10.0	0.75

	icon
0	rain
1	rain
2	rain
3	rain
4	snow
...	...
2673	rain
2674	partly-cloudy
2675	partly-cloudy
2676	rain
2677	clear

[2678 rows x 20 columns]

```
[ ]: date_column = weather_df_allegHENY2.pop('date')
weather_df_allegHENY2.insert(0, 'date', date_column)
```

```
[ ]: weather_df_allegHENY2
```

```

[ ]:      date      name tempmax tempmin temp feelslikemax \
0  2017-01-01 Allegheny County 43.9 31.4 37.1 40.1
1  2017-01-02 Allegheny County 46.3 37.0 42.6 43.3
2  2017-01-03 Allegheny County 48.6 42.9 45.1 46.3
3  2017-01-04 Allegheny County 44.3 21.1 33.8 37.4
4  2017-01-05 Allegheny County 22.6 19.0 20.6 20.8
...      ...      ...      ...      ...      ...
2673 2024-04-27 Allegheny County 79.1 53.0 65.3 79.1
2674 2024-04-28 Allegheny County 82.0 63.2 72.6 81.5
2675 2024-04-29 Allegheny County 82.9 62.6 74.1 82.1
2676 2024-04-30 Allegheny County 72.3 61.1 65.9 72.3
2677 2024-05-01 Allegheny County 79.9 53.7 67.5 79.9

      feelslikemin feelslike dew humidity precip precipcover preciptype \
0      29.0      33.2 28.3 71.1 0.025 12.50 rain,snow
1      33.2      38.4 38.9 87.1 0.095 29.17 rain
2      37.7      41.6 42.6 91.0 0.793 70.83 rain
3      8.0      23.6 23.4 67.4 0.052 29.17 rain,snow
4      7.0      11.2 11.5 68.4 0.056 25.00 snow
...      ...      ...      ...      ...      ...
2673      53.0      65.3 50.3 60.5 0.090 25.00 rain
2674      63.2      72.6 55.3 55.6 0.000 0.00 rain
2675      62.6      73.9 54.6 52.0 0.000 0.00 rain
2676      61.1      65.9 56.1 71.0 0.014 4.17 rain
2677      53.7      67.5 48.6 55.4 0.000 0.00 none

      snow snowdepth cloudcover uvindex severerisk moonphase \
0      0.0      0.2 41.7 4 0.0 0.12
1      0.0      0.0 87.9 1 0.0 0.16
2      0.0      0.0 78.7 0 0.0 0.19
3      0.0      0.0 73.0 4 0.0 0.23
4      0.2      0.0 83.5 3 0.0 0.25
...      ...      ...      ...      ...      ...
2673 0.0 0.0 65.9 7 10.0 0.63
2674 0.0 0.0 47.9 10 10.0 0.67
2675 0.0 0.0 27.2 9 10.0 0.70
2676 0.0 0.0 68.1 6 10.0 0.74
2677 0.0 0.0 16.2 8 10.0 0.75

      icon
0      rain
1      rain
2      rain
3      rain
4      snow
...      ...
2673      rain

```

```

2674 partly-cloudy
2675 partly-cloudy
2676         rain
2677         clear

```

[2678 rows x 20 columns]

```

[ ]: merged_df = pd.merge(weather_df_allegheny2, overdose_cleaned, on='date',
    ↪how='inner')
merged_df

```

```

[ ]:
      date      name  tempmax  tempmin  temp  feelslikemax \
0  2018-01-07 Allegheny County    26.2    -4.2  10.1         18.1
1  2018-01-09 Allegheny County    35.8    28.0  32.5         34.7
2  2018-01-14 Allegheny County    15.3     2.5   9.2         14.6
3  2018-01-14 Allegheny County    15.3     2.5   9.2         14.6
4  2018-01-14 Allegheny County    15.3     2.5   9.2         14.6
...
5308 2024-04-13 Allegheny County    59.2    44.5  51.1         59.2
5309 2024-04-13 Allegheny County    59.2    44.5  51.1         59.2
5310 2024-04-13 Allegheny County    59.2    44.5  51.1         59.2
5311 2024-04-14 Allegheny County    81.1    45.0  64.1         79.9
5312 2024-04-14 Allegheny County    81.1    45.0  64.1         79.9

      feelslikemin  feelslike  dew  humidity  ...  age_range  race \
0          -16.6         0.6  -3.8        57.2  ...    30 - 39  BLACK
1           24.1        26.8  26.6        78.9  ...    40 - 49  WHITE
2           -6.0         3.5   0.5        68.0  ...    40 - 49  WHITE
3           -6.0         3.5   0.5        68.0  ...    20 - 24  WHITE
4           -6.0         3.5   0.5        68.0  ...    30 - 39  WHITE
...
5308          37.4        47.9  33.0        51.5  ...    20 - 24  WHITE
5309          37.4        47.9  33.0        51.5  ...    20 - 24  WHITE
5310          37.4        47.9  33.0        51.5  ...    20 - 24  WHITE
5311          39.9        62.9  39.5        42.6  ...    20 - 24  WHITE
5312          39.9        62.9  39.5        42.6  ...    30 - 39  BLACK

      ethnicity_desc  accidental_exposure  victim_od_drug_id \
0              Unknown                      N              3288
1  NON-HISPANIC OR NOT LATINO                      N              527
2  NON-HISPANIC OR NOT LATINO                      N              800
3  NON-HISPANIC OR NOT LATINO                      N              804
4  NON-HISPANIC OR NOT LATINO                      N              799
...
5308  NON-HISPANIC OR NOT LATINO                      N          55826
5309  NON-HISPANIC OR NOT LATINO                      N          55825
5310  NON-HISPANIC OR NOT LATINO                      N          55825

```

5311	NON-HISPANIC OR NOT LATINO	N	55849
5312	NON-HISPANIC OR NOT LATINO	N	55827

	susp_od_drug_desc	naloxone_administered	administration_id	survive \
0	HEROIN	N	0	Y
1	HEROIN	Y	244	Y
2	HEROIN	Y	454	Y
3	HEROIN	Y	457	Y
4	HEROIN	Y	452	Y
...	...	...	...	...
5308	HEROIN	Y	25971	Y
5309	FENTANYL	Y	25971	Y
5310	FENTANYL	Y	25972	Y
5311	HEROIN	Y	25981	Y
5312	HEROIN	Y	25973	U

	dose_unit
0	0
1	0
2	0
3	0
4	0
...	...
5308	4
5309	4
5310	4
5311	4
5312	4

[5313 rows x 38 columns]

```
[ ]: merged_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5313 entries, 0 to 5312
Data columns (total 38 columns):
#   Column              Non-Null Count  Dtype
---  -
0   date                 5313 non-null  datetime64[ns]
1   name                 5313 non-null  object
2   tempmax              5313 non-null  float64
3   tempmin              5313 non-null  float64
4   temp                 5313 non-null  float64
5   feelslikemax         5313 non-null  float64
6   feelslikemin         5313 non-null  float64
7   feelslike            5313 non-null  float64
8   dew                  5313 non-null  float64
9   humidity             5313 non-null  float64
```

```

10 precip                5313 non-null    float64
11 precipcover           5313 non-null    float64
12 preciptype            5313 non-null    object
13 snow                  5313 non-null    float64
14 snowdepth             5313 non-null    float64
15 cloudcover            5313 non-null    float64
16 uvindex               5313 non-null    int64
17 severerisk            5313 non-null    float64
18 moonphase             5313 non-null    float64
19 icon                  5313 non-null    object
20 dose_count            5313 non-null    object
21 incident_id           5313 non-null    object
22 incident_time         5313 non-null    object
23 day                   5313 non-null    object
24 incident_county_name  5313 non-null    object
25 incident_state        5313 non-null    object
26 victim_id            5313 non-null    object
27 gender_desc           5313 non-null    object
28 age_range             5313 non-null    object
29 race                  5313 non-null    object
30 ethnicity_desc        5313 non-null    object
31 accidental_exposure   5313 non-null    object
32 victim_od_drug_id     5313 non-null    object
33 susp_od_drug_desc     5313 non-null    object
34 naloxone_administered 5313 non-null    object
35 administration_id     5313 non-null    object
36 survive               5313 non-null    object
37 dose_unit             5313 non-null    object
dtypes: datetime64[ns](1), float64(15), int64(1), object(21)
memory usage: 1.5+ MB

```

```
[ ]: merged_df.to_csv('merged_df.csv', index=False)
```

```
[ ]: data_path = os.path.join(os.path.expanduser("~"), "Desktop", "merged_df.csv")
```

```
[ ]: merged_df.to_csv(data_path, index=False)
```

```
[ ]:
```